

## ORIGINAL ARTICLE

# Intraspecific trait variation of carrion beetle species and communities across elevations

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## Abstract

1. Filtering processes across environmental gradients can structure patterns of trait variation within communities. The community-weighted mean (CWM) is a metric that is commonly used to indicate the directionality of such filtering processes and the optimal adaptive strategy of taxa within community. Proximity to the CWM indicates higher fitness, and deviations from this optimal value result in changes in the relative abundances of coexisting species.
2. We investigated patterns of intraspecific trait variation in four coexisting carrion beetle species (Coleoptera: Silphidae) across elevational gradients. The study was conducted in temperate forest ecosystems with distinct natural vegetation zones ranging from 950 m to 1700 m above sea level.
3. Of the 12 traits measured, we found that intraspecific variation ranged from 34% (body length) to 100% (ratio of elytra length, head length and head width to body length) and accounted for a larger proportion of variation than interspecific variation in 7 traits. For most traits, trait range, which indicates the niche breadth of species at a given elevation, was positively correlated with relative abundance. The CWMs of traits associated with long-distance dispersal decreased with elevation, whereas those associated with microhabitat use showed the opposite trend. Soil temperature influenced tibia length after controlling for the effects of species identity, but soil water content had no effect on trait variation. Patterns of variation

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in body width and thorax width of two beetle species supported the CWM-optimality hypothesis; however, patterns of trait variation in body width and thorax length of another species showed an opposite trend.

4. Our study identifies several traits that can be highly variable within carrion beetle species. Such high levels of intraspecific trait variation may enable populations to adapt across a wide range of elevations and vegetation types.

#### KEYWORDS

adjustment, biodiversity distribution, Coleoptera, functional trait, species coexistence, temperate mountain landscape

## INTRODUCTION

Traits influence organismal performance and determine how individuals interact with the environment and with each other (McGill et al., 2006). Studying traits rather than species alone provides a mechanistic understanding of how organisms respond to environment and how they will respond to future environmental change (Kellermann & van Heerwaarden, 2019; Weemstra et al., 2021). Community trait variation can be attributed to interspecific and intraspecific variation (Ackerly, 2003). While interspecific trait variation is the product of macroevolutionary processes (Anderegg, 2023), intraspecific trait variation (ITV) is the result of adaptive microevolution, phenotypic plasticity or sexual dimorphism (Anderegg, 2023; Bolnick et al., 2011; Moran et al., 2016). The amount of intraspecific variation has been shown to be comparable to interspecific variation in plants (He et al., 2018; Messier et al., 2010; Siefert et al., 2015) and vertebrates (Munsee et al., 2021; Xavier Jordani et al., 2019). Due to the significant contribution of intraspecific component to overall trait variation, it has been suggested that community ecology should shift its focus from species to traits, with an emphasis on ITV (de Bello et al., 2021; Messier et al., 2010; Westerband et al., 2021).

ITV quantifies how different the trait is between coexisting individuals of the same species (Albert et al., 2011). It can be attributed to variation between individuals and between sexes (DeLong et al., 2021). Common measure of ITV are coefficient of variation (CV) and trait niche breadth (range or width). Coefficient of variation is dimensionless, making it comparable across traits, species and studies (Yang et al., 2020). Trait niche breadth compares the trait spectrum of local populations across large environmental gradients and can be used as a proxy for local adaptive potential (Maccagni & Willi, 2022). Populations with higher ITV are considered to be better equipped to thrive in a wider range of environments (Muscarella & Uriarte, 2016). Species exhibiting higher ITV are likely to have higher relative abundance in community due to their ability to adapt to wider environmental conditions. When comparing traits of different functions, those with higher ITV will vary more between communities, particularly for those species adapted to environments across large spatial gradients (Siefert et al., 2015).

How well a species performs in local community depends not only on the breadth of functional traits and environmental determinants but also on the trait deviation from the community mean, which

indicates the environmental optimum for that trait (Lisner et al., 2021). The community-weighted mean (CWM) has often been used to indicate the optimal adaptive strategy of taxa in local environmental conditions and the degree of directionality of environmental filtering processes (DeMalach et al., 2022; Muscarella & Uriarte, 2016). In a given community, dominant species tend to occupy core positions in the trait space and vary less around optimal values (Umaña et al., 2015). Species with trait values closest to the CWM values are expected to have the highest fitness, that is, the 'CWM-optimality' hypothesis (Muscarella & Uriarte, 2016). Indeed, traits defining multidimensional niche space enable different species to coexist in local communities across extensive geographic or environmental gradients (Lourenço et al., 2020; Wong et al., 2019). Adaptive strategies and environmental filtering of organisms along gradients can be inferred from both CWM and trait variance partitioned across different ecological scales, especially within and between species (Messier et al., 2010; Moran et al., 2016; Umaña & Swenson, 2019).

Mountains provide an excellent natural platform to study environmental adaptation through trait means and variation (Alexander et al., 2018; Chamberlain et al., 2020). This is particularly valuable for assessing the concept of optimality, as elevation ranges provide diverse environmental conditions over short distances (Rahbek et al., 2019). Such conditions allow the study of trait adaptability without interference of populations. Due to high ITV, trait CWMs have been shown to change with elevation in several taxa. For example, elevation gradients affect several functional traits in plants by altering temperature and water availability in the environment (Vitasse et al., 2014; Walck et al., 2011). In endotherms, the morphology of several traits changes in different directions with elevation (Luzuriaga-Aveiga & Weir, 2019). However, ectotherms such as insects, which constitute a large proportion of terrestrial arthropods, have been less studied.

Beetles (Insecta: Coleoptera) comprise about 25% of the known animal species on Earth. Diverse morphologies and complex feeding habits allow them to adapt to different environments and contribute to ecological functions such as phytophagy, pollination, energy transfer, nutrient cycling and decomposition of vertebrates and other organic matter (Fountain-Jones et al., 2015; Ślipiński & Lawrence, 2019). Along elevational gradients, beetles typically exhibit morphological changes (Stanbrook et al., 2021). For example, body size of the

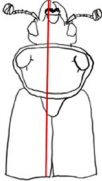
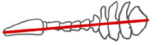
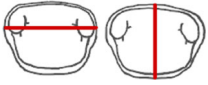
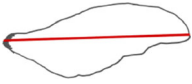
carrion beetle *Silpha carinata* Herbst, 1783 has been shown to decrease with elevation but without change in ITV within populations (Baranovská & Knapp, 2018; Wong et al., 2019). In addition, flight performance of beetles is usually reduced at high elevations; changes in wing or body shape may compensate for the reduction in air density (Hodkinson, 2005). Therefore, each beetle species may have a different temperature tolerance and adapt differently to environmental conditions through different mechanisms via functional traits (DeCastro-Arrazola et al., 2023; Giménez Gómez et al., 2022; Williamson et al., 2022).

Carrion beetles (Coleoptera: Silphidae) contribute significantly to the functioning of mountain ecosystems by recycling nutrients from vertebrates across elevations. They feed on vertebrate carcasses and also prey on snails, earthworms and small insects (Ikeda et al., 2008; Tsai et al., 2020). Due to their distinctive functions, Silphidae are emerging as a model organism in animal trait ecology (Fountain-Jones et al., 2015; Tsai et al., 2020; Wong et al., 2019). In this study, we examined variation in 12 morphological traits of carrion beetles distributed across elevations from 950 to 1700 m above sea level (a.s.l.)

(Table 1). Hindwing and leg are associated with individual long- and short-distance dispersal ability, respectively, while body length, width, head and thorax may be related to microhabitat use. In contrast, antennae and elytra may help individuals avoid and protect themselves from predators (Fountain-Jones et al., 2015). Although most of these traits are diagnostic, they are likely to show some variation due to some ecological processes, making them suitable for revealing environmental filtering and justifying the functionality of the traits for adaptation across elevations.

We first partitioned variation in the traits with different putative functions into five hierarchical components, that is, elevation, plot, interspecies and within-species (including sex and inter-individual variation), and compared the contributions of these ecological scales among different functional traits. We hypothesised that traits with different functions would show different degrees of ITV. We then tested whether individual trait variation was influenced by elevation, soil temperature and water content. As species with high ITVs are likely to tolerate a wide range of environmental conditions and thus thrive in local communities (Shen et al., 2019), we hypothesised

TABLE 1 Morphological traits of carrion beetles measured in this study.

| Trait                                             | Measurement position                                                                | Ecological functions                                       | Effect (E)/<br>response (R) trait | Reference                                      |
|---------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------|-----------------------------------|------------------------------------------------|
| Body length                                       |   | Fecundity, microhabitat use                                | E and R                           | Nichols et al. (2009), Slade et al. (2007)     |
| Body width                                        |  | Microhabitat use                                           | R                                 | Gibb et al. (2006)                             |
| Antenna length                                    |  | Habitat preference, hunting capability, predator detection | E and R                           | Bauer et al. (1998), Talarico et al. (2007)    |
| Head (width and length)                           |  | Microhabitat use                                           | R                                 | Ribera et al. (1999)                           |
| Thorax (width and length)                         |  | Microhabitat use                                           | R                                 | Fountain-Jones et al. (2015)                   |
| Elytra length                                     |  | Parasite resistance, protection from predation             | E and R                           | Asgari et al. (2021), Goczał and Beutel (2023) |
| Hindwing (maximal length)                         |  | Long-distance dispersal                                    | E and R                           | Barton et al. (2011)                           |
| Leg (femur length, tibia length and total length) |  | Short-distance dispersal                                   | R                                 | Barton et al. (2011)                           |

positive correlations of trait CVs and niche breadth with relative abundance of species. Finally, as predicted by the CWM-optimality hypothesis, we examined which of the 12 morphological traits were responsible for carrion beetle performance, as represented by relative abundances, across elevations.

## MATERIALS AND METHODS

### Study area

The study area is located on the northern slope of the Changbai Mountain ( $127^{\circ}43'–128^{\circ}16' E$ ;  $41^{\circ}41'–42^{\circ}51' N$ ; Figure 1a), one of the most intact and biodiverse forest ecosystems in temperate regions (Stone, 2006). It experiences a dry and windy spring, a short and wet summer, a cool autumn with widespread fog and a long and cold winter (Zhang et al., 2018). Mean annual temperature ranges from  $-4.17$  to  $2.71^{\circ}C$  across elevations, and mean annual precipitation ranges from 683 to 955 mm (Xie et al., 2022). Vegetation is characterised by a distinctive elevation zonation from mixed coniferous and broad-leaved forests below 1100 m, mixed coniferous forests between 1100 and 1500 m, subalpine mixed coniferous forests between 1500 and 1800 m, birch forests between 1800 and 2100 m and alpine tundra with dwarf trees and shrubs above 2100 m (Figure S1) (Bai et al., 2011; Zhang et al., 2016).

### Collection of species and trait data

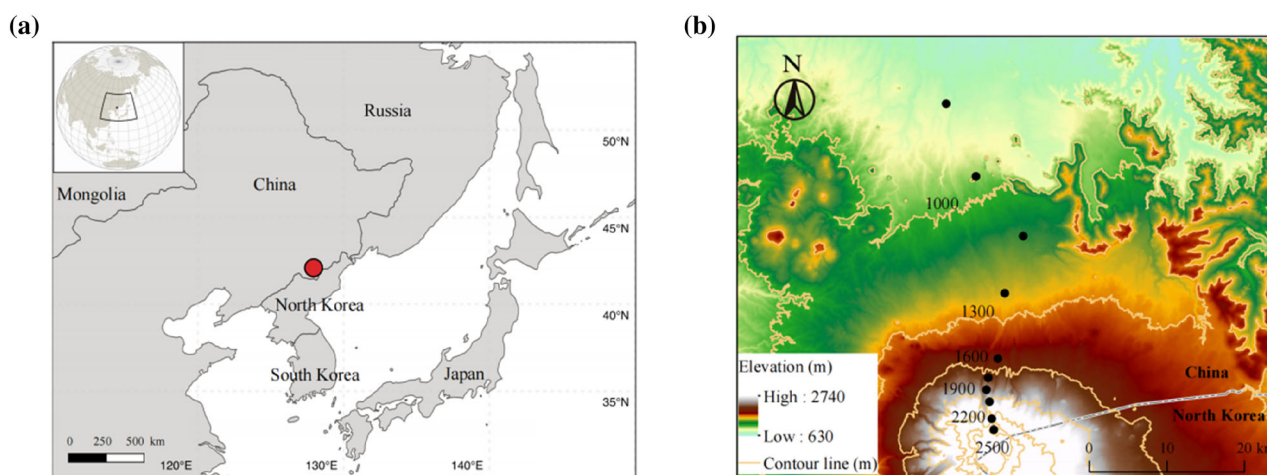
Beetle samples were collected at 10 elevations: 800, 950, 1100, 1250, 1400, 1550, 1700, 1850, 2150 and 2600 m a.s.l. (Figure 1b) in July 2020, as adult activity in the mountains typically begins in late June, persists throughout July, and declines by late August (Smith et al., 2000). At each elevation, seven plots were selected between

100 and 200 m apart to reduce spatial autocorrelation between communities (Gao et al., 2018). Individuals of Silphidae were collected using carrion-baited pitfall traps (Baz et al., 2007) constructed from 450 mL plastic cups (9 cm high, 7.5 cm diameter). Each trap was filled with 150 mL of saturated salt solution and left in the field for 72 h. Trapped animals were preserved in ethanol at  $-20^{\circ}C$ . Silphidae individuals were identified to species level using the key 'The Carrion Beetles of China (Coleoptera: Silphidae)' (Ji, 2012) and were verified by mitochondrial cytochrome c oxidase subunit I (COI) sequences.

A maximum of five males and five females of each Silphidae species collected from each plot were randomly selected for morphological trait measurements. If less than 10, all individuals were measured. Twelve morphological traits were measured under a binocular microscope with the Nikon NIS-Elements D 5.01.00 (Table 1). Except for body length, the other traits were calculated as a ratio to body length due to potential isometric scaling effects.

### Environmental data

As temperature and water availability can vary with elevation, and for most biota these two factors are the main drivers of community composition across large environmental gradients, soil temperature and water content were used as potential drivers of carrion beetle traits. Annual mean soil temperature for the 10 elevations was extracted from Global Soil Bioclimatic variables at a resolution of  $30 \times 30$  arcseconds (Lembrechts et al., 2022). Soil temperature at the studied elevations was highly correlated with air temperature (correlation coefficient = 0.96;  $p < 0.001$ ; Xie et al., 2022). As carrion beetles of our study system live near the ground surface with a close affinity to the soil habitat, soil temperature rather than air temperature was tested for its effects on carrion beetle traits (Lembrechts et al., 2022). Soil water content was derived from the same study plots on the Changbai Mountain (Lux et al., 2022).



**FIGURE 1** Geographical position (a; red dot) and topology with the 10 sampling elevations from 800 m to 2600 m above sea level (b; black dots) of Changbai Mountain in northeast China. At each elevation, carrion beetle communities were sampled from seven plots at 100–200 m intervals.

## Statistical analysis

To make the results comparable between species and between elevations, only those species that were distributed and co-occurred in at least six consecutive elevations were included in trait analyses. All analyses were performed in R 4.1.2 ([www.r-project.org](http://www.r-project.org)).

Before trait analyses, outlier measurements were removed using the 'EnvStats' package (Millard, 2013). Principal component analysis (PCA) was applied to the 12 traits, and the first four PCs were retained in the trait variance partitioning analysis. Variance of each trait and trait PCs was partitioned into five nested hierarchical scales including (1) elevation, (2) plot, (3) species, (4) sex and (5) inter-individual. Inter-individual and sex variations of a species were considered as ITV. To determine which of the nested ecological scales contributed most to the variation in traits with different functions, linear mixed effects models were applied using the packages 'nlme' and 'ape', with elevation, plot, species and sex as nested random factors as  $\sim 1|Elevation/Plot/Species/Sex$  (Messier et al., 2010) using the function *lme*.

Several linear mixed effects models were built to search for the best environmental predictors of trait values among soil temperature, soil water content and elevation. The environmental factors were set as fixed factors and species as a random factor, and all possible combinations of the environmental factors were included in the models (Table S2). The support of the models was compared using the Akaike Information Criterion corrected for small sample size. A linear model with no fixed factors was also included for comparison.

As the plot contributed virtually no trait variation, trait CV was calculated for each species at each elevation. The trait niche breadth for each species at each elevation was calculated as the ratio of the species trait range at a given elevation (i.e., maximum value minus minimum value of a trait at a given elevation) to the full range of trait values across all elevations (i.e., maximum value minus minimum value of a trait across all elevations; Fajardo & Siefert, 2019). The relative abundance of species in the community at each elevation was

correlated with CV and niche breadth using Pearson correlation with significance test.

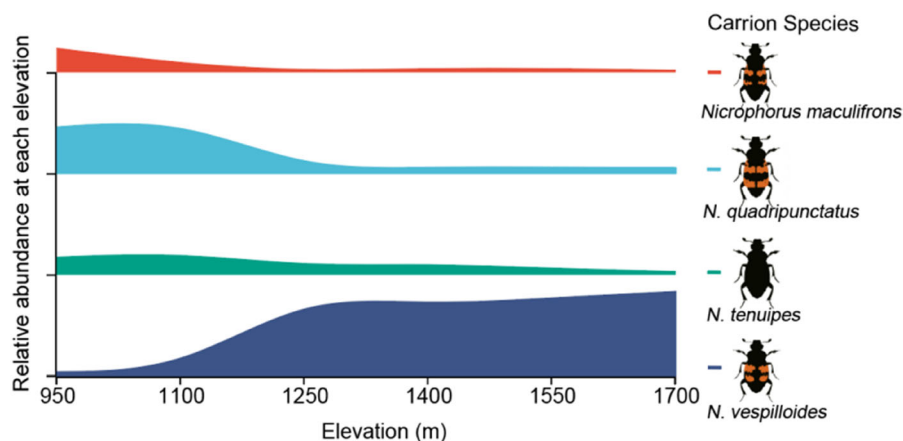
For each trait, CWM was calculated at each elevation using the 'FD' package (Violle et al., 2007). Species trait means were also calculated for each elevation. Both CWM and species trait means were modelled by linear regression with elevation as a fixed predictor. To test the 'CWM-optimality' hypothesis, trait  $\Delta CWM$  was calculated following Muscarella and Uriarte (2016). As not all individuals were empirically measured, the trait values of those without measurements were inferred from the trait probability distribution, given the mean and standard deviation of the trait in the community at a given elevation, using the 'TPD' package. The trait  $\Delta CWM$  of each species was modelled by linear regression with species log relative abundance as a fixed predictor.

## RESULTS

Eight carrion beetle species (1034 individuals) were collected across the elevations, with the four species, *Nicrophorus maculifrons* Kraatz, 1877, *N. quadripunctatus* Kraatz, 1877, *N. tenuipes* Lewis, 1887 and *N. vespilloides* Herbst, 1783, coexisting from 950 m to 1700 m a.s.l. and used as the focus for trait analyses (815 individuals). The relative abundance of each of the four *Nicrophorus* species along the elevation gradient is shown in Figure 2. For the four species, a total of 297 individuals were measured for the 12 morphological traits (Table S1).

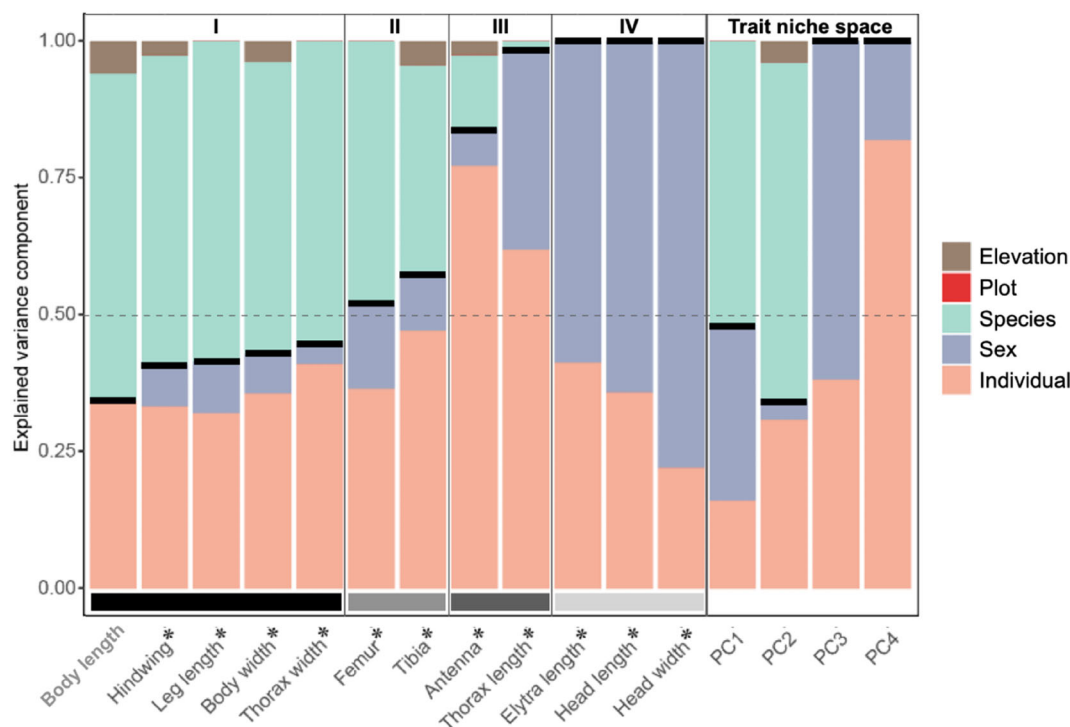
### Trait variation at different scales

The PC1 axis of the trait space (Figure S2) was mainly associated to head width, leg length and thorax length. Individual variation along this axis was similarly contributed by both intraspecific and interspecific variation (49.2% and 51.8%, respectively). Individual scores on



**FIGURE 2** Relative abundance of each of the four *Nicrophorus* species along the elevational gradient in the Changbai Mountain. While the relative abundance of *N. maculifrons*, *N. quadripunctatus* and *N. tenuipes* decreased with elevation, *N. vespilloides* increased.





**FIGURE 3** Percentage of trait variance explained by nested hierarchical scales: Interindividual (between individuals of one sex within a species), sex (males versus females within a species), species (different species within a plot), plot (seven plots at one elevation) and elevation (six elevations within the study area). According to the contribution of the scales, traits are assigned to different types, I, II, III and IV. Type I: Interspecific variation contributes more than intraspecific variation; Type II: Interspecific and intraspecific variation contribute equally; Type III: Intraspecific variation contributes more than interspecific variation and inter-individual differences than sex differences; Type IV: All variation is caused by intraspecific variation, with sex being a more important factor than inter-individual differences. The plot scale contributed virtually no trait variation. The position of the thick black line between the species- and sex-contributed variation in each bar indicates the summed proportion of intraspecific variation. The niche space of multidimensional morphological traits was constructed using principal component (PC) analysis, and individual scores along the first four axes were used in variance partitioning. Asterisk indicates ratio of trait to body length. Dashed line indicates 50% of trait variance. See text in Results section for details.

the PC2 axis, correlated with body length, thorax width and body width, were contributed more by species (61.9%) than by ITV (34.2%), with elevation contributing a small proportion of variation (3.9%). Along the PC2, the species *N. maculifrons* had the highest scores, with body length and hindwing being longer than that of the other three species. Although trait variation on PC1 and PC2 was mainly contributed by species differences (Figure S2a), there was some overlap between species in trait space. In contrast, variation on PC3 and PC4 was contributed exclusively by sex and individual differences (Figure 3; Figure S2b).

In 7 of the 12 traits examined, ITV (sex + individual) contributed more than interspecific variation (Types II, III and IV). Variation in elytra length, head length and head width was almost entirely contributed by ITV, and traits differed more between sexes than within sex (Type IV). Antenna and thorax length were also largely contributed by ITV, but within-sex variation was greater than between-sex variation, while species and elevation were less important (Type III). Intraspecific variation in femur and tibia length contributed about 50%, with some species variation (Type II). By contrast, for body length, hindwing, leg length, body width and thorax width, interspecific variation contributed more than ITV (Type I). Of the traits examined, elevation alone contributed little to variation and was only found in five traits: body length,

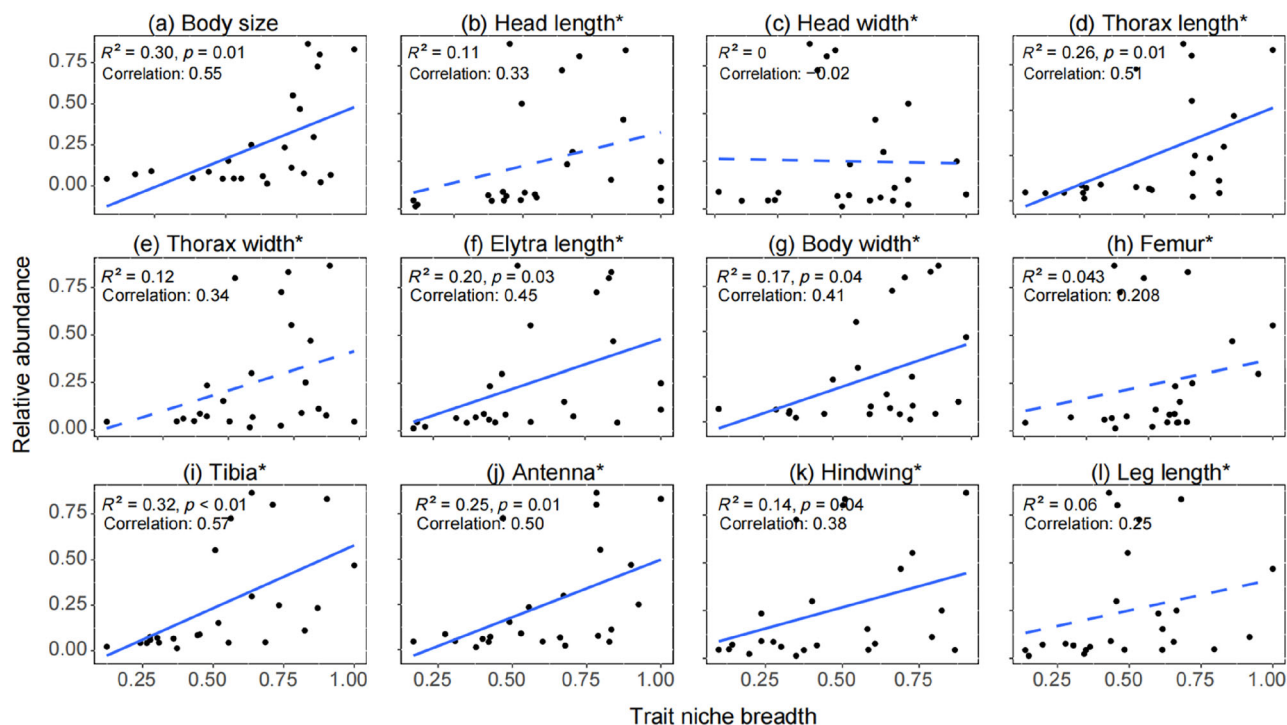
body width, tibia, antenna and hindwing, while plots of the same elevation did not contribute to any of the traits examined (Figure 3).

### Relationship between trait CV and breadth with species relative abundance

Except for hindwing, trait CV showed no correlation with species relative abundance (Figure S3), but trait breadth did (Figure 4). The breadth of 7 of the 12 traits, including body length, thorax length, elytra length, body width, tibia, antenna and hindwing, were significantly positively correlated with species relative abundance.

### Environmental gradient of community and species traits

From low to high elevations, CWM of body length, head width, elytra length and hindwing decreased significantly (Figure 5a,c,f,k). However, CWM of thorax length, thorax width and body width increased significantly (Figure 5d,e,g). Within each species, there was at least one trait that showed ITV associated with elevation. Body length of



**FIGURE 4** Relationship between trait elevational niche breadth of carrion beetles and their relative abundance in Changbai Mountain. Asterisk indicates ratio of trait to body length. Blue solid line indicates significant linear relationship ( $p < 0.05$ ).

*N. vespilloides*, head length of *N. tenuipes*, femur of *N. quadripunctatus* and hindwing of *N. maculifrons* decreased significantly with elevation (Figure 5a,b,h,k). Elytra length of *N. tenuipes*, tibia of *N. maculifrons* and *N. vespilloides* and antenna and leg length of *N. maculifrons* significantly increased with elevation (Figure 5f,i,j,l).

Soil temperature was the best predictor for tibia (correlation coefficient:  $-0.58$ ,  $p = 0.02$ ), but not for the other traits. Soil water content had no effect on trait variation (Table S2).

## CWM-optimality hypothesis

Four of the 12 traits examined showed either supportive or opposing patterns as suggested by the CWM-optimality hypothesis, including body length, thorax length, thorax width and body width (Figure 6). Species relative abundances were negatively correlated with  $\Delta$ CWM of body length, thorax width and body width in *N. tenuipes* and thorax width and body width in *N. quadripunctatus* (Figure 6a,c,d), supporting the CWM-optimality hypothesis. In contrast, the  $\Delta$ CWM of thorax length and body width in *N. vespilloides* was positively correlated with relative abundance (Figure 6b,d), contrary to the hypothesis. However, *N. maculifrons* showed no significant patterns (Figure 6a–d).

## DISCUSSION

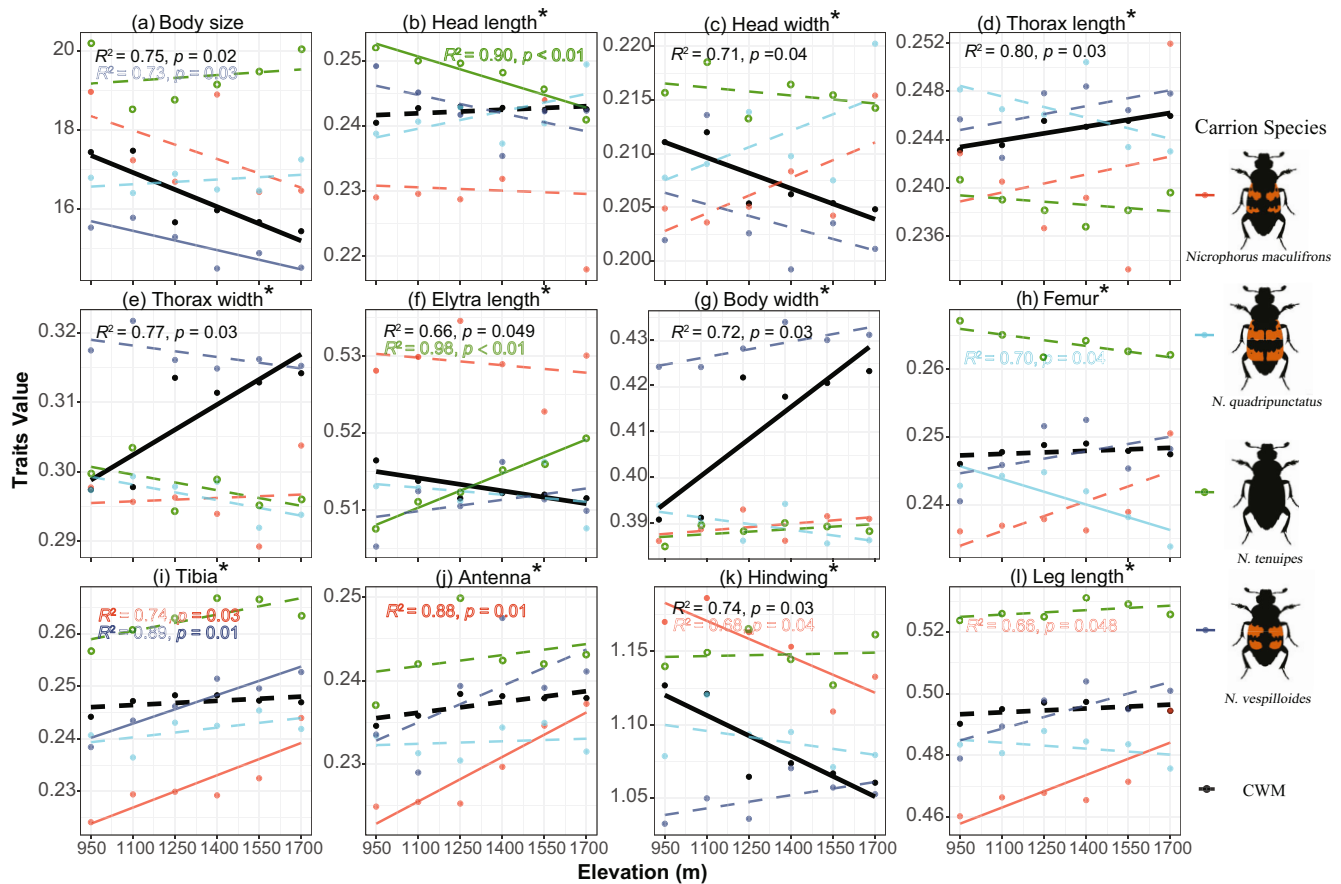
Using carrion beetles collected across elevations, our study demonstrated the contributions of different ecological scales (i.e., intraspecific

between sexes and between individuals, interspecific between coexisting species within an elevation and elevational difference independent of species) to the variation of different functional traits. We showed how trait variation may have influenced species fitness, as represented by the relative abundance of the four coexisting model scavenger species.

## ITV as a prerequisite for adaptation

Our study highlighted the contribution of ITV to total trait variation, which appeared to be important for of the 12 traits examined. A study examining morphological traits in African mountain ants found that ITV contributed only 1%–4% of the total variation in Weber's length, mandible length, eye position and leg length (Gaudard et al., 2019). In dung beetles, ITV accounted for 4% to 6% of trait variation in dry body mass and hind leg length in Amazonian forests (Griffiths et al., 2016). However, our study showed that ITV contributed at least 34% of the total variation in the traits studied, highlighting the prerequisites of ITV for carrion beetle adaptation through traits with different functions: short- and long-distance dispersal, microhabitat use and predator avoidance.

ITV is composed of sex differences and inter-individual variation. Significant morphological differences between sexes have been reported in stag, rhinoceros and dung beetles (Ślipiński & Lawrence, 2019). While Collard et al. (2021) reported sexual dimorphism in body size of some *Nicrophorus* species, our study further showed that females had longer elytra and hindwing lengths than



**FIGURE 5** Relationship between elevation and community-weighted mean (CWM; black dots and line) of trait and species trait mean (coloured dots and lines) of the four carrion beetle species across the six elevations of Changbai Mountain. Asterisk indicates ratio of trait to body length. Solid line indicates significant linear relationship ( $p < 0.05$ ).

males, suggesting that females are better at self-protection and long-distance dispersal than males in this mountain system. Studies in other arthropods, such as ants, spiders and praying mantises, have demonstrated the role of females in reproduction, offspring protection and resource acquisition and their impact on population wealth and fitness (Barry et al., 2008; Toft & Albo, 2015). Similarly, in the mountain system studied, females are likely to be the key to population adaptation to new environments, as their elytra and hindwing traits are critical for survival and colonisation in changing habitats.

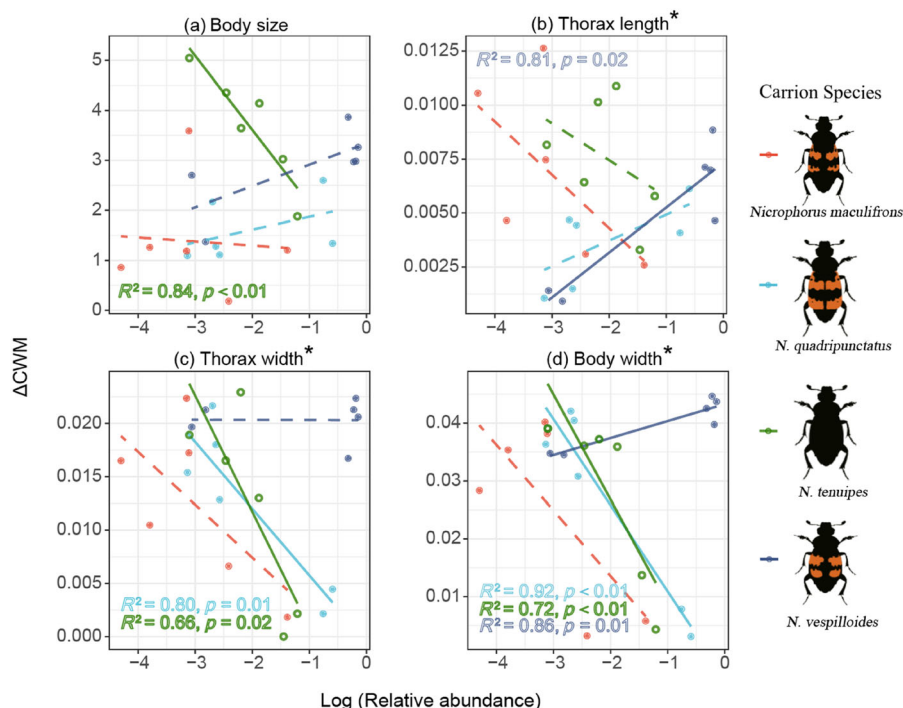
Inter-individual variation contributed more than 50% to antenna and thorax length—two traits indicative of habitat preference and microhabitat use (Fountain-Jones et al., 2015; Gibb et al., 2006). Similarly, considerable ITV has been demonstrated across elevations for plant leaf traits, carbon, nitrogen and stable isotope content (Albert et al., 2010; Umaña & Swenson, 2019). When plant traits (such as elemental or stable isotope content) and insect traits (such as antenna and thorax length) are brought into analogy, the large contribution of intraspecific trait variability suggests its general importance for the adaptation, function and responses of biota across taxa and even kingdoms to environmental change (Violle et al., 2012).

### Functional trait variation for adaptation in carrion beetles across elevations

Our study used two ITV measures: CV and niche breadth. Compared with CV, niche breadth was a better predictor of the relative abundance of species at a given elevation, as there were significant, albeit weak, correlations between the relative abundance and niche breadth for of the 12 traits examined in our study. Given the adaptability of arthropods to local environments, along elevation gradients the same species will undergo corresponding trait changes to adapt to different local environments (Messier et al., 2017). Across the six elevations characterised by different vegetation zones, only CV in hindwing size, as a proxy for flight ability, was significantly positively correlated with species relative abundance. Variation in flying traits is important for competition and dispersal (Ikeda et al., 2008; Keena, 2018). Our result suggests that hindwing is a critical trait for adaptation of carrion beetles to mountain systems.

Niche breadth is a key factor in determining species occupancy and local abundance in a given region, as species with a broader niche are likely to have higher occupancy and abundance (Heino, 2005). Species with higher relative abundance in a community tend to have a





**FIGURE 6** Relationship between log-transformed relative abundance and distance of the species trait mean at an elevation to the community-weighted mean ( $\Delta\text{CWM}$ ) of the four carrion beetle species in Changbai Mountain. Asterisk indicates ratio of trait to body length. Solid line indicates significant linear relationship ( $p < 0.05$ ).

competitive advantage and are likely to be plastic in resource acquisition (Ashton et al., 2010). While most of the traits examined in this study support this hypothesis, niche breadth of head length and width and thorax width were not correlated with species relative abundance. The size of the head and thorax of ground beetles has been used to indicate their microhabitat niche, as these morphological traits influence their ability to navigate through different substrate types and access various resources (Fountain-Jones et al., 2015). The non-significant results of our study suggest that while these traits are important for microhabitat use, their variation is more closely related to environmental changes rather than directly influencing species relative abundance. In contrast, the relative abundance of species was positively correlated with thorax length, and variation in this trait was driven more by inter-individual variation than the other ecological scales (Type III).

Femur and tibia are part of the leg and can be used as a proxy for dispersal ability (Barton et al., 2011). Our study shows that tibia supports the niche breadth hypothesis, but femur does not. This is likely because the muscles required for locomotion are based on the tibia. In addition, tibia is usually expanded towards the apex, where it often bears combs of spines, which enables certain functions such as antenna cleaning or hearing (Lawrence & Britton, 2019). The traits with positive correlations between niche breadth and species relative abundance suggest that the ITV of this morphological (niche) dimension is relevant to carrion beetle adaptation across elevations.

## Elevational filtering processes via functional traits

Environmental filtering revealed by trait CWM can be contributed by two variables: the mean trait values of species in different environments and changes in the relative abundance or turnover of species in local communities (Lajoie & Vellend, 2015). We defined the community as the four focal species coexisting at all six elevations. All observed variations in trait CWM were due to changes in either relative abundance or ITV, or both, but not to species turnover. Although CWM is a common indicator of environmental filtering, effects of environmental factors can vary between species, resulting in inconsistencies between CWM and species mean trait values.

At the community level, 7 of the 12 traits showed significant elevational patterns, either negative or positive correlations, suggesting that environmental filtering processes are at work and that these traits are functional for carrion beetles. Body size, as represented by body length in our study, is a comprehensive indicator of individual response to environmental change. The decrease in body length of carrion beetle community with elevation was found in our study. *Silpha carinata* and *S. longicornis* of the same carrion beetle family have been shown to have the same pattern as ours (Baranovská & Knapp, 2018; Nishikawa et al., 2010). A decrease in the mean body size of populations with an increase in elevation has also been found in carabid and dung beetles (Stanbrook et al., 2021; Sukhodolskaya & Ananina, 2015). The decrease pattern of *N. vespilloides* in our study is likely due to a reduction in the length of the season suitable for beetle

growth and activity at higher elevations, or in the quality and availability of food at different elevations (Nishikawa et al., 2010; Stillwell et al., 2007). In contrast, the CWM of body width and thorax width increased significantly with elevation, suggesting that these traits enhance carrion beetles' thermoregulation, energy storage and burrowing capabilities, thereby maximising their microhabitat use efficiency at high elevations (Barton et al., 2011).

CWM of hindwing size of carrion beetles decreased significantly with elevation in our study system. A reduction in flight function at higher elevations is a common phenomenon in mountain insects (Hill et al., 2021; Hodkinson, 2005; Lawrence & Britton, 2019). Although hindwings are important for long-distance dispersal in beetles, this trait can also be influenced by habitat structure (Bui et al., 2020). For geometrid moths, warming-induced range shifts to higher elevations have been reported for smaller species, which show a reduction in community mean wing size at high elevations (Wu et al., 2019). In our study, the change in hindwing CWM was due to size reduction of a single species, *N. maculifrons*, and its decrease in relative abundance. Indeed, *N. maculifrons* was the species with the largest hindwing of the four species studied, but its hindwing size was smaller than that of *N. quadripunctatus* at the two highest elevations. Limb morphology, as a proxy for short-distance dispersal, did not differ across elevations at the community level. However, at the species level, a positive correlation was observed between elevation and tibia length in *N. maculifrons* and *N. vespilloides*. This suggests that species-specific ITV in dispersal traits may contribute to the adaptation of certain species to higher elevations, although not all species follow this trend. However, this pattern was offset by a decrease in the relative abundance of species within the local community.

Our study showed that only the short-distance dispersal traits were affected by soil temperature, particularly tibia length. In addition to its function in individual movement, this trait may also aid in thermoregulation (Krasnov et al., 1996). However, it is also likely that we have not captured the most important drivers of variation in carrion beetle traits within species, as it has been reported that air temperature fluctuation determines beetle growth and development (Hofmann et al., 2024; Lamb & Gerber, 1985; McMillan et al., 2005). Our study indicated no significant relationship between trait variation and soil water content. The general climate of the study area is characterised by low temperatures but relatively high rainfall and humidity. Variations in water availability therefore do not appear to stress carrion beetles (Yu et al., 2011; Zhang et al., 2022). While our results provide insight into the relationships between traits and the environment, further research is needed to fully understand the complexity of trait variation within carrion beetle communities.

## CWM-optimality hypothesis and species coexistence processes

High levels of ITV clearly enable populations to adapt across diverse elevations and vegetation types (Anderegg, 2023; Laiolo & Obeso, 2015). Our study is the first to test the CWM-optimality

hypothesis in animals. A previous study showed that 70% of tree species supported the hypothesis with at least one trait, while 25% of species rejected it (Muscarella & Uriarte, 2016). Our study demonstrated that the four carrion beetle species responded differently to elevations. Two species supported the CWM-optimality hypothesis, one opposed it, and one neither supported nor opposed. The traits supporting and opposing the hypothesis were all related to microhabitat use rather than dispersal, indicating that microhabitat properties are key factors influencing carrion beetle communities, although beetles are thought to be good dispersers.

If  $\Delta$ CWM trait values are negatively related to the relative abundance (supporting the CWM-optimality hypothesis), this implies that the species have optimal functional strategies given adapted to local abiotic conditions. Positive relationships (opposing the CWM-optimality hypothesis) indicate niche differentiation (Muscarella & Uriarte, 2016). For example, body length of *N. tenuipes* was identified as a trait filtered by the abiotic environment. The low dominance of this species explains its high  $\Delta$ CWM of body length across all elevations. Microhabitat use by *N. quadripunctatus* and *N. tenuipes* is driven by abiotic environmental filtering through body and thorax width, while *N. vespilloides* utilises niche differentiation in thorax length and body width, particularly at low elevation when coexisting with the other species. At higher elevations, however, *N. vespilloides* was better adapted than the other three species, contributing to the positive CWM-elevation relationships due to its higher relative abundance. Despite surviving at high elevations, the other three species were less adapted than *N. vespilloides* but coexisted through niche differentiation in microhabitat use. These contrasting patterns suggest that carrion beetles adapt to elevation by different mechanisms.

Our findings highlight that microhabitat properties, such as variations in vegetation density, substrate composition and microclimatic conditions, differ along the studied gradient and require different traits in carrion beetles for effective adaptation. For instance, beetles at lower elevations may require traits for navigating denser vegetation, whereas those at higher elevations may need traits for coping with cooler temperatures and sparser vegetation. As trait niche breadth increases with relative abundance, such intraspecific variation provides a basis for coexistence of different species across elevations, resulting in the overall CWM-environment relationships.

## Study limitations and future research directions

Although our study was conducted in July when carrion beetles seem to be the most active in a year, their true diversity in the study region may be underestimated. Future studies need to include variation of traits across seasons, to reveal the temporal dynamics of community assembly processes of carrion beetles, which are mediated by functional traits. In addition to mean annual soil temperature and water content, the inclusion of other climate variables at different temporal scales would also help to predict intraspecific trait changes of carrion beetles, particularly for those species endemic to mountainous regions, which is more vulnerable to climate change.

Other scavenging insects with similar trophic guilds may co-occur with carrion beetles, but use resources at different time. Flies are among the first to discover carcasses and begin to feed and lay eggs on them. As carrion beetles and dung beetles arrive, the resources available to flies reduce rapidly. Beetles may also ingest fly eggs or larvae when feeding on carrion (Charabidze et al., 2021). Furthermore, the sub-social behaviour of carrion beetles is likely to give them a competitive advantage over dung beetles (Barton & Bump, 2019). These complex trophic interactions between different scavenger insects require further investigation. In addition to the morphological traits examined in our study, trophic traits and phenology are likely to provide insights into both intraspecific and interspecific niche partitioning of scavengers to reveal their community assembly processes across elevations.

## CONCLUSION

Our study suggests the central role of ITV in adaptation of carrion beetles, highlighting its potential role in short- and long-distance dispersal, microhabitat use and predator avoidance. Functional trait variation, particularly hindwing CV and tibia niche breadth, emerges as a key mechanism influencing species abundance at different elevations, suggesting the importance of dispersal ability for carrion beetle adaptation in mountain ecosystems. Our exploration provides insights into elevational filtering processes and underscores the different strategies used by carrion beetle species to coexist at different elevations. This study represents one of the few instances where the CWM-optimality hypothesis has been tested in animals. Our research advances understanding of the complex dynamics of functional trait variation and adaptation strategies of beetles in response to environmental changes, paving the way for future studies on ecological processes of insect communities across large environmental gradients.

## AUTHOR CONTRIBUTIONS

**Qiao-Qiao Ji:** Conceptualization; data curation; investigation; formal analysis; funding acquisition; methodology; project administration; software; resources; validation; visualization; writing – original draft; writing – review and editing. **Zhijing Xie:** Writing – original draft; writing – review and editing; formal analysis. **Yunga Wu:** Writing – review and editing; investigation. **Zhuoma Wan:** Investigation; writing – review and editing. **Caiyi Xu:** Project administration; resources; writing – review and editing. **Donghui Wu:** Conceptualization; funding acquisition; project administration; resources; supervision; writing – review and editing. **Ting-Wen Chen:** Conceptualization; writing – original draft; methodology; validation; visualization; writing – review and editing; software; formal analysis; data curation; supervision. **Alejandro Ordonez:** Conceptualization; formal analysis; methodology; software; supervision; writing – original draft; writing – review and editing.

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## CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

## DATA AVAILABILITY STATEMENT

The data sets analysed during the current study are available in the Zenodo repository at <https://doi.org/10.5281/zenodo.10430284>.

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## SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

**Figure S1.** Elevation zonation of vegetation on Changbai Mountain in July: (a) mixed coniferous and broad-leaved forests below 1100 m, (b) mixed coniferous forests between 1100 and 1500 m, (c) subalpine mixed coniferous forests between 1500 and 1800 m, (d) birch forests between 1800 and 2100 m, (e) alpine tundra with dwarf trees and shrubs above 2100 m, (f) Tianchi crater with volcanic rocks.

**Figure S2.** Principal component analysis of 12 morphological traits of individuals of four carrion beetle species, *Nicrophorus maculifrons*, *N. quadripunctatus*, *N. tenuipes* and *N. vespilloides*, distributed from 950 m to 1700 m a.s.l. on Changbai Mountain. Asterisk indicates ratio of trait to body length.

**Figure S3.** Relationship between coefficient of variation (CV) of carrion beetle traits and their relative abundance in Changbai Mountain. Asterisk indicates ratio of trait to body length. Blue solid line indicates significant linear relationship ( $p < 0.05$ ).

**Table S1.** Number of individuals (Ind; measured/collected) and trait values (mean±SD) of the four co-occurring *Nicrophorus* carrion beetle species at each of the six elevations (Elev.) in Changbai Mountain. Asterisk indicates ratio of trait to body length.

**Table S2.** Akaike Information Criterion corrected for small sample size (AICc) of linear mixed effects models for variation in carrion beetle traits as predicted by soil temperature (ST), soil water content (WC) and elevation without (denoted by +) and with interaction (denoted by \*) between factors. Carrion beetle species was included as a random factor. The best model for trait variation is shown in bold with the lowest AICc.

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